

Duality III: Marginal Costs and Dual Simplex

An optimal dual solution explains the effect of a changed requirement. Dual simplex repairs a basis when that change makes a primal quantity negative.

Our route: marginal costs $\rightarrow$ dual feasibility $\rightarrow$ dual-simplex pivots $\rightarrow$ geometry and degeneracy.

Bertsimas and Tsitsiklis §§4.4–4.5.

Marginal costs

The dual simplex method

Two pivots and their geometry

Complete calculations and proofs

The quantities and their costs are already familiar

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

The optimal quantities are $x^* = (1, 0, 1)$, with cost 19.

The two equality right-hand sides are requirements that must be matched exactly.

Now ask: what happens to the best cost when one requirement changes?

Recall the optimal dual variables

$$\begin{aligned} \max \quad & 8p_1 + 3p_2 \\ \text{s.t.} \quad & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.} \end{aligned}$$

p_1 corresponds to the first equality and p_2 to the second.

$$p^* = (2, 1), \quad 8(2) + 3(1) = 19.$$

The identity $13x_1 + 10x_2 + 6x_3 = 2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) + 7x_2$ proves the matching bound.

Increase the first requirement from 8 to 9

Keep $x_2 = 0$, and solve the changed equations:

$$3x_1 = 3 \Rightarrow x_1 = 1, \quad 5(1) + 3x_3 = 9 \Rightarrow x_3 = \frac{4}{3}.$$

$$13(1) + 6\left(\frac{4}{3}\right) = 21 = 19 + 2(1).$$

The same dual variables give the lower bound $2(9) + 1(3) = 21$. This feasible solution attains it, so the cost really increases by 2.

The first price $p_1 = 2$ measures cost per unit increase in requirement 1.

Increase the second requirement from 3 to 3.6

Keep the first requirement at 8 and again set $x_2 = 0$:

$$\begin{aligned} 3x_1 &= 3.6 \Rightarrow x_1 = \frac{6}{5}, \\ 5\left(\frac{6}{5}\right) + 3x_3 &= 8 \Rightarrow x_3 = \frac{2}{3}. \end{aligned}$$

$$13\left(\frac{6}{5}\right) + 6\left(\frac{2}{3}\right) = 19.6 = 19 + 1(0.6).$$

The dual lower bound is also $2(8) + 1(3.6) = 19.6$. Thus $p_2 = 1$ is the cost per extra unit of requirement 2 here.

A price prediction is exact while the basis stays feasible

If the right-hand side changes from b to $b + \Delta b$, and the same optimal basis stays feasible, then

$$v(b + \Delta b) = v(b) + (p^*)^\top \Delta b.$$

Changing only the right-hand side leaves the dual constraints and the reduced costs unchanged.

The original dual variables still give a valid lower bound after any right-hand-side change. Equality with the new optimum needs a feasible primal solution that attains that bound.

For a nondegenerate optimal basis, sufficiently small changes in any direction preserve feasibility.

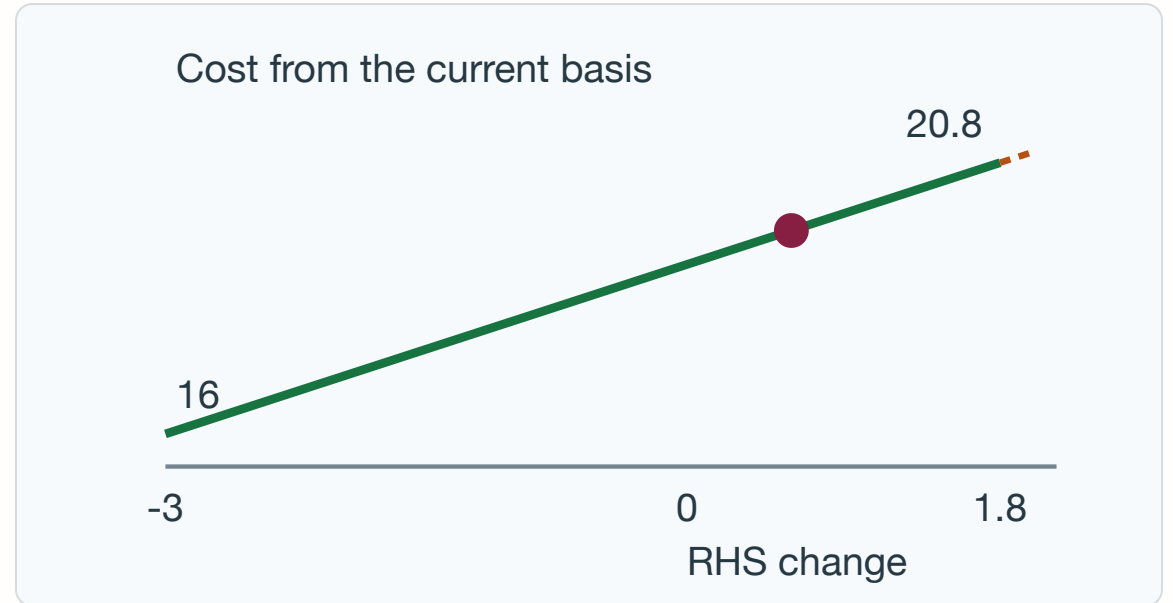
How far can we keep the same quantities pattern?

Replace 3 by $3 + \varepsilon$ in the second equality, keeping $x_2 = 0$.

$$x_1 = 1 + \frac{\varepsilon}{3}, \quad x_3 = 1 - \frac{5\varepsilon}{9}.$$

$$v = 19 + \varepsilon \quad \text{while} \quad -3 \leq \varepsilon \leq \frac{9}{5}.$$

Shown: a change of 0.6 keeps both basic quantities positive.



Change 0.6: quantities (1.2, 0, 0.667); cost 19.6.
The current basis remains feasible and optimal.

A negative quantity ends this basis's prediction

At $\varepsilon = 2$, the second requirement is 5. The old pattern gives

$$x_1 = \frac{5}{3}, \quad x_2 = 0, \quad x_3 = -\frac{1}{9}.$$

This is infeasible. Its predicted cost 21 is not the cost of a feasible solution.

The dual variables $(2, 1)$ still certify the lower bound $2(8) + 5 = 21$. We must reoptimize to find the actual minimum.

Optional: solve the new optimum explicitly.

Interpret the price with the constraint in view

Equality requirements: a price is the change in minimum cost per unit increase in a required total. Equality dual variables can be positive, zero, or negative.

Range: the prediction holds while a suitable optimal basis remains feasible. At a change of basis, the marginal cost can change.

At degeneracy, prices may not be unique and the value function may have a corner. A dual variable still gives a valid bound.

We will calculate allowable changes systematically in sensitivity analysis.

Interpret the columns as ingredients and the dual variables as prices

Think of each column as the contents of one unit of an ingredient. The two equality rows specify the required contents of the final mixture.

Ingredient	Contents A_j	Cost c_j	Value $(p^*)^\top A_j$
1	(5, 3)	13	$2(5) + 1(3) = 13$
2	(1, 1)	10	$2(1) + 1(1) = 3$
3	(3, 0)	6	$2(3) + 1(0) = 6$

Dual feasibility says the value assigned to an ingredient's contents cannot exceed its purchase cost.

Used ingredients are priced exactly at their cost

$$c - (A^\top p^*) = (0, 7, 0), \quad x^* = (1, 0, 1).$$

Ingredients 1 and 3 are used, and their price–cost gaps are zero. Ingredient 2 costs 7 more than the value assigned to its contents, so it is unused.

$$13x_1 + 10x_2 + 6x_3 = 19 + 7x_2.$$

This is the dictionary objective from simplex. Its reduced cost 7 measures the extra cost of using ingredient 2 after adjusting the basic ingredients.

Zero gap permits, but does not force, positive use. The mixture also attains its dual value: $(p^*)^\top b = 19$.

The mixture problem behind the picture

Column	B	C	D	E	F
First total	1/2	3	0	2	0
Second total	1/2	0	3	1/2	0
Cost	7	4	4	1	4

$$\begin{aligned} \min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\ \text{s.t.} \quad & \frac{1}{2}x_B + 3x_C + 2x_E = 1, \\ & \frac{1}{2}x_B + 3x_D + \frac{1}{2}x_E = 1, \\ & x_B + x_C + x_D + x_E + x_F = 1, \quad x \geq 0. \end{aligned}$$

Each x_j is a mixture weight. The two required totals are $(1, 1)$; the third equality makes the weights sum to 1. Thus a dual bound uses **three** row weights.

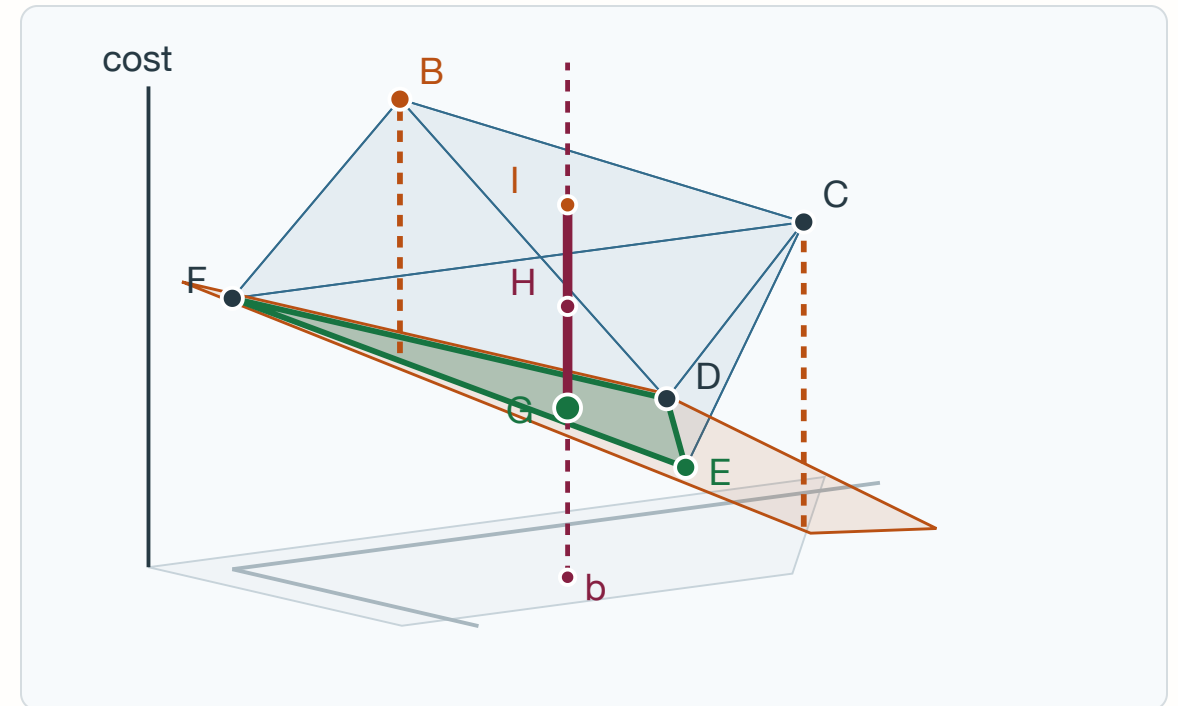
Weight the three requirements to get a lower bound

Use $p = (-3/2, 0, 4)$. For a column with entries (a_1, a_2) , its weighted value is the plane height $z = 4 - \frac{3}{2}a_1$.

The constant 4 weights the sum-to-one equality.

Every column lies on or above this plane. At the required totals, it gives the bound $-\frac{3}{2}(1) + 0(1) + 4(1) = \frac{5}{2}$.

The mixture $(x_D, x_E, x_F) = (1/4, 1/2, 1/4)$, with the others zero, attains $5/2$.



Weighted column bounds certify the mixture

$A_j = (a_{1j}, a_{2j}, 1)$ includes normalization. Use $p = (-3/2, 0, 4)$.

$$c_j \geq p^\top A_j = -\frac{3}{2}a_{1j} + 0a_{2j} + 4 \quad \text{for every column } j.$$

Multiply by $x_j \geq 0$, add, and use the three required totals:

$$\begin{aligned} c^\top x &\geq -\frac{3}{2} \sum_j a_{1j}x_j + 0 \sum_j a_{2j}x_j + 4 \sum_j x_j \\ &= -\frac{3}{2}(1) + 0(1) + 4(1) = \frac{5}{2}. \end{aligned}$$

D, E, F attain the bound. B and C have dual slacks (reduced costs) $15/4$ and $9/2$.

Dual simplex: keep the bound, repair the quantities

We continue with minimization. A changed requirement can make a basic quantity negative while the reduced costs remain nonnegative.

The old basis still supplies a feasible dual solution and a lower bound. Dual simplex changes the basis until its primal quantities become feasible too.

Primal simplex preserves primal feasibility. Dual simplex preserves dual feasibility.

Bertsimas and Tsitsiklis §4.5. We will first pivot equations, then organize the same arithmetic in a tableau.

One basis determines both x and p

$$\min c^\top x \quad \text{s.t. } Ax = b, \ x \geq 0; \quad \max b^\top p \quad \text{s.t. } A^\top p \leq c.$$

p_i is the dual variable for equality i ; p is unrestricted. Assume A has independent rows.

Choose independent columns B , in the same order as the basic variables. Set the nonbasic variables to zero and solve

$$Bx_B = b, \quad B^\top p = c_B.$$

Here c_B lists their objective coefficients. The associated dual slacks are the reduced costs: $\bar{c}_j = c_j - A_j^\top p$.

Two separate tests on the same dictionary

Property	Test for a minimization dictionary
Primal feasibility	All basic values are nonnegative: $B^{-1}b \geq 0$.
Dual feasibility	All reduced costs are nonnegative: $\bar{c}_j \geq 0$.
Both	The associated x and p are optimal.

$$b^\top p = c_{\mathcal{B}}^\top B^{-1}b.$$

Before primal feasibility is restored, this common algebraic value is a dual lower bound. It is not the cost of a feasible primal solution.

Example 4.7: a dual-feasible starting dictionary

$$\begin{array}{ll}\min & f = 2x_1 + 6x_2 + 10x_3 \\ \text{s.t.} & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \quad x \geq 0.\end{array}$$

Use basic variables (x_4, x_5) and set $x_1 = x_2 = x_3 = 0$.

$$x_4 = 2, \quad x_5 = -1, \quad p = (0, 0).$$

The primal basic solution is infeasible. But p satisfies every dual inequality, since the reduced costs are $(2, 6, 10, 0, 0) \geq 0$.

Start with equations: which variable must leave?

$$\begin{aligned}x_4 &= 2 + 2x_1 - 4x_2 - x_3, \\x_5 &= -1 - 4x_1 + 2x_2 + 3x_3, \\f &= 2x_1 + 6x_2 + 10x_3.\end{aligned}$$

$x_5 = -1$ violates nonnegativity, so choose x_5 to leave the basis.

In its equation, increasing x_2 or x_3 can raise x_5 . Increasing x_1 makes the violation worse.

Two entering choices are possible. We must choose one that keeps every objective coefficient nonnegative.

Compare the two candidates before pivoting

Candidate	Positive coefficient in the x_5 equation	Reduced cost / coefficient
x_2	2	$6/2 = 3$
x_3	3	$10/3$

Choose the smaller ratio: x_2 enters and x_5 leaves.

Why? Solving the pivot row for x_2 subtracts 3 times its coefficients from the objective coefficients. The coefficient of x_3 becomes $10 - 3(3) = 1 \geq 0$.

Choosing x_3 would instead make the coefficient of x_2 equal to $6 - 2(10/3) = -2/3$, losing dual feasibility.

Pivot step 1: solve the selected equation

$$x_5 = -1 - 4x_1 + 2x_2 + 3x_3.$$

$$x_2 = \frac{1}{2} + 2x_1 - \frac{3}{2}x_3 + \frac{1}{2}x_5.$$

The new nonbasic variables are x_1, x_3, x_5 . At zero, the new basic variable is $x_2 = 1/2$.

Now substitute this equation into the other basic row and the objective, exactly as in primal simplex.

Pivot step 2: update the other row and objective

$$\begin{aligned}x_4 &= 2 + 2x_1 - 4\left(\frac{1}{2} + 2x_1 - \frac{3}{2}x_3 + \frac{1}{2}x_5\right) - x_3 \\&= -6x_1 + 5x_3 - 2x_5, \\f &= 2x_1 + 6\left(\frac{1}{2} + 2x_1 - \frac{3}{2}x_3 + \frac{1}{2}x_5\right) + 10x_3 \\&= 3 + 14x_1 + x_3 + 3x_5.\end{aligned}$$

All new reduced costs are nonnegative. The dual lower bound increases from 0 to 3.

Both tests now pass: the pivot found an optimum

$$x_2 = \frac{1}{2} + 2x_1 - \frac{3}{2}x_3 + \frac{1}{2}x_5,$$

$$x_4 = -6x_1 + 5x_3 - 2x_5,$$

$$f = 3 + 14x_1 + x_3 + 3x_5.$$

$$x = (0, \frac{1}{2}, 0, 0, 0), \quad f = 3.$$

In tableau row order the basic variables are (x_4, x_2) . All basic values and reduced costs are nonnegative.

$x_4 = 0$ is basic: the primal solution is degenerate, but the dictionary still certifies optimality.

Find the associated dual variables explicitly

$$\begin{aligned} \min \quad & f = 2x_1 + 6x_2 + 10x_3 \\ \text{s.t.} \quad & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \quad x \geq 0. \end{aligned}$$

The new basic columns are $A_4 = (1, 0)^\top$ and $A_2 = (4, -2)^\top$. Set their dual constraints active:

$$4p_1 - 2p_2 = 6, \quad p_1 = 0 \quad \implies \quad p = (0, -3).$$

$$c - A^\top p = (14, 0, 1, 0, 3) \geq 0, \quad b^\top p = 2(0) + (-1)(-3) = 3.$$

Write the same dictionary as a tableau

Move nonbasic terms to the left in constraint rows. Write the objective as $-f + 2x_1 + 6x_2 + 10x_3 = 0$.

Row	Column 0	x_1	x_2	x_3	x_4	x_5
R_0	0	2	6	10	0	0
x_4	2	-2	4	1	1	0
x_5	-1	4	-2	-3	0	1

Column 0 is the RHS in constraint rows and $-f_0$ in R_0 , where $f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j$. Columns always follow natural variable order.

The same pivot is three row operations

In the preceding tableau, R_1 is the x_4 row and R_2 is the x_5 row. First normalize the pivot row:

$$R'_2 = -\frac{1}{2} R_2, \quad R'_1 = R_1 - 4R'_2, \quad R'_0 = R_0 - 6R'_2.$$

Row	Column 0	x_1	x_2	x_3	x_4	x_5
R_0	−3	14	0	1	0	3
x_4	0	6	0	−5	1	2
x_2	1/2	−2	1	3/2	0	−1/2

Keep the constraint-row positions: the former x_5 row now represents x_2 . The upper-left entry is −3, so the objective value is 3.

The dual ratio test in dictionary notation

Suppose the chosen negative row and objective are

$$x_k = \beta + \sum_{j \in \mathcal{N}} \alpha_j x_j, \quad \beta < 0; \quad f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j, \quad \bar{c}_j \geq 0.$$

An entering variable must have $\alpha_j > 0$. Choose

$$j \in \arg \min_{i \in \mathcal{N}: \alpha_i > 0} \frac{\bar{c}_i}{\alpha_i}.$$

x_k leaves; x_j enters. In a tableau, the same row coefficient is $v_j = -\alpha_j < 0$, so the ratio is $\bar{c}_j/|v_j|$.

The smallest ratio preserves every dual inequality

After solving the selected row for x_j , its reduced cost is eliminated. Let $t = \bar{c}_j/\alpha_j \geq 0$.

New coefficient of $x_i = \bar{c}_i - t\alpha_i$, new coefficient of $x_k = t$.

Other coefficient	Why it stays nonnegative
$\alpha_i > 0$	$t \leq \bar{c}_i/\alpha_i$ by the minimum ratio.
$\alpha_i \leq 0$	$\bar{c}_i - t\alpha_i \geq \bar{c}_i \geq 0$.

This is why taking any variable with a helpful sign is not enough. The objective row determines how far the dual variables may change.

One iteration: diagnose, choose, pivot, check

Step	Operation
1. Check the start	All reduced costs must be nonnegative.
2. Choose a leaving row	If every basic value is nonnegative, stop: optimal. Otherwise choose a negative basic value.
3. Choose an entering variable	Among positive RHS dictionary coefficients in that row, take the smallest reduced-cost/coefficient ratio.
4. Pivot and repeat	Solve for the entering variable, substitute everywhere, and check the new basic values.

If there is no eligible entering variable, the constraints are infeasible. We explain that case next.

A row that cannot be repaired proves infeasibility

Suppose the selected row is

$$x_k = -1 - 2x_2 - 3x_4, \quad x_k, x_2, x_4 \geq 0.$$

The right side is at most -1 for every allowed choice. It cannot equal a nonnegative variable.

In general: $\beta < 0$ and every $\alpha_j \leq 0$ imply $x_k = \beta + \sum_j \alpha_j x_j < 0$.

The primal is infeasible. Because this is a dual-feasible dictionary, the dual is unbounded above; the appendix constructs its improving ray.

Example 4.8: follow two pivots in both spaces

$$\min f = x_1 + x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \quad x_1, x_2 \geq 0.$$

$$\max 2p_1 + p_2 \quad \text{s.t.} \quad p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \quad p_1, p_2 \geq 0.$$

p_1, p_2 correspond to the two primal $\geq$ constraints. Add surplus variables $x_3, x_4 \geq 0$:

$$x_1 + 2x_2 - x_3 = 2, \quad x_1 - x_4 = 1.$$

The surplus-column dual inequalities $-p_i \leq 0$ are exactly $p_i \geq 0$. The paired plots use the same letter for a basis in both spaces.

Example 4.8: basis A starts dual feasible

$$x_3 = -2 + x_1 + 2x_2,$$

$$x_4 = -1 + x_1,$$

$$f = x_1 + x_2.$$

$$\mathcal{B} = (x_3, x_4), \quad x = (0, 0, -2, -1), \quad p = (0, 0), \quad f_0 = 0.$$

Choose x_3 to leave. The ratios are $1/1 = 1$ for x_1 and $1/2$ for x_2 , so x_2 enters.

Example 4.8: first pivot gives basis B

From $x_3 = -2 + x_1 + 2x_2$, solve for x_2 and substitute:

$$x_2 = 1 - \frac{1}{2}x_1 + \frac{1}{2}x_3,$$

$$x_4 = -1 + x_1,$$

$$f = 1 + \frac{1}{2}x_1 + \frac{1}{2}x_3.$$

$$\mathcal{B} = (x_2, x_4), \quad x = (0, 1, 0, -1), \quad p = (1/2, 0), \quad f_0 = 1.$$

Reduced costs remain nonnegative. But $x_4 = -1$, so we must pivot again.

Example 4.8: second pivot gives basis C

Only x_1 can increase $x_4 = -1 + x_1$. Let x_1 enter and x_4 leave; substitute $x_1 = 1 + x_4$.

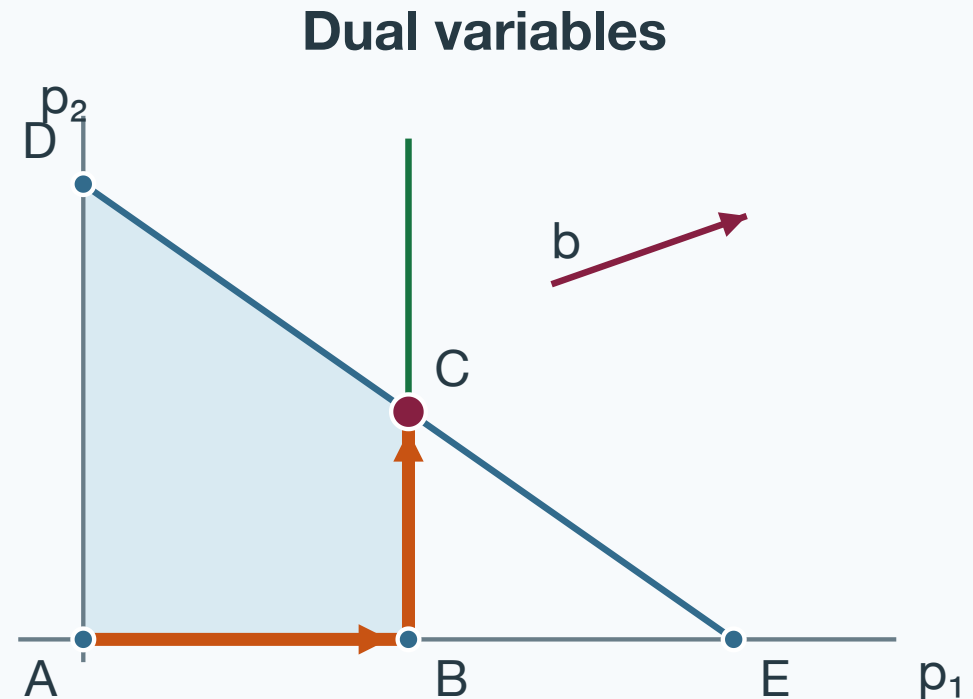
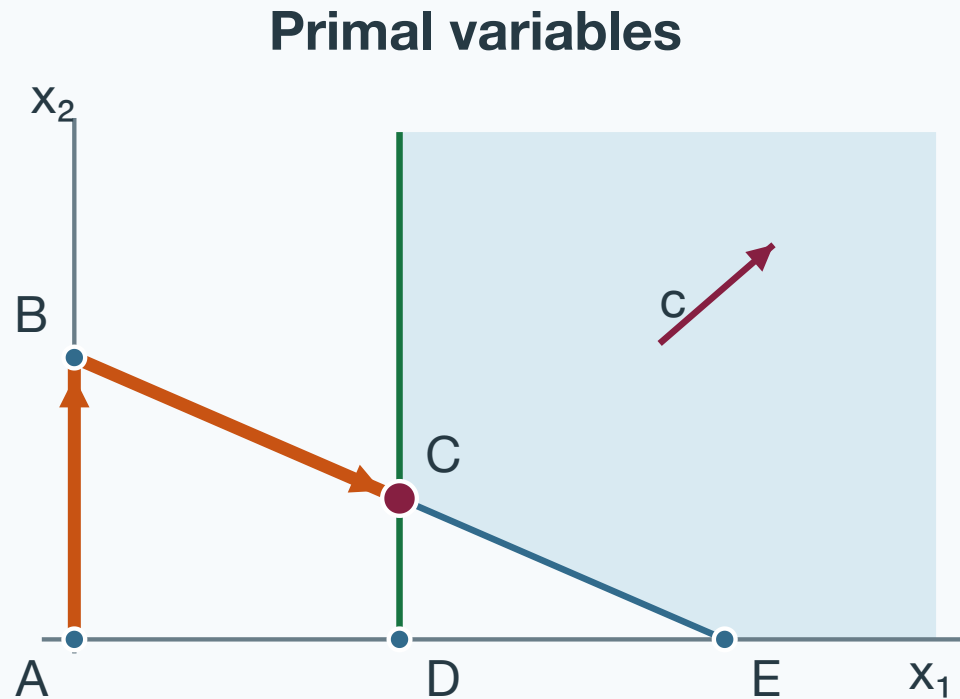
$$\begin{aligned}x_1 &= 1 + x_4, \\x_2 &= \frac{1}{2} + \frac{1}{2}x_3 - \frac{1}{2}x_4, \\f &= \frac{3}{2} + \frac{1}{2}x_3 + \frac{1}{2}x_4.\end{aligned}$$

$$\mathcal{B} = (x_1, x_2), \quad x = (1, 1/2, 0, 0), \quad p = (1/2, 1/2), \quad f_0 = 3/2.$$

Every basic value and reduced cost is nonnegative. Both primal and dual solutions are optimal.

The primal path is infeasible until its last point

Example 4.8: $\min x_1 + x_2$, with $x_1 + 2x_2 \geq 2$, $x_1 \geq 1$, $x \geq 0$. The dual maximizes $2p_1 + p_2$, with $p_1 + p_2 \leq 1$, $2p_1 \leq 1$, $p \geq 0$.



Orange: the same basis changes in both spaces. Maroon arrows show increasing objective. Shading is clipped by the plot window.

The same basis explains both plotted points

Label	Basic variables	Primal point (x_1, x_2)	Dual point (p_1, p_2)
A	(x_3, x_4)	$(0, 0)$	$(0, 0)$
B	(x_2, x_4)	$(0, 1)$	$(1/2, 0)$
C	(x_1, x_2)	$(1, 1/2)$	$(1/2, 1/2)$
D	(x_1, x_3)	$(1, 0)$	$(0, 1)$
E	(x_1, x_4)	$(2, 0)$	$(1, 0)$

The algorithm used $A \rightarrow B \rightarrow C$. E is dual infeasible and D is primal infeasible. At C, both tests pass.

Adjacent dual bases share one active constraint here. In higher dimensions, consecutive bases share all but one of their basic-column equalities.

Why it is called dual simplex

In dual space, the algorithm moves between adjacent feasible basic solutions and increases the dual objective.

The same basis changes appear in a dictionary of the primal equations, even while their basic values are negative.

It is not necessary to convert the dual into standard form and solve that new tableau. Such a conversion can produce a different sequence of pivots.

Dual simplex works directly with the original basis and its associated dual constraints.

Return to the changed requirement in Example 4.6

$$\begin{array}{ll}\min & f = 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \quad x \geq 0.\end{array}$$

Keep the previous basis (x_1, x_3) and solve its two equations:

$$x_1 = \frac{5}{3} - \frac{1}{3}x_2, \quad x_3 = -\frac{1}{9} + \frac{2}{9}x_2, \quad f = 21 + 7x_2.$$

The reduced cost is still 7, but $x_3 < 0$. Dual simplex can repair the old basis without restarting Phase I.

One pivot repairs the changed requirement

In the preceding dictionary, x_3 leaves and x_2 enters. Solve $x_3 = -1/9 + 2x_2/9$ for x_2 :

$$\begin{aligned}x_1 &= \frac{3}{2} - \frac{3}{2}x_3, \\x_2 &= \frac{1}{2} + \frac{9}{2}x_3, \\f &= \frac{49}{2} + \frac{63}{2}x_3.\end{aligned}$$

$$x^* = (3/2, 1/2, 0), \quad f^* = 49/2 = 24.5.$$

The old bound was 21. The repaired basis raises the bound to the attainable minimum 24.5.

Practice: choose the pivot and write the new dictionary

Minimize f subject to the dictionary equations and $x_1, x_2, x_3, x_4 \geq 0$:

$$x_3 = 4 + x_1 - 2x_2,$$

$$x_4 = -2 + x_1 + 4x_2,$$

$$f = 3x_1 + 2x_2.$$

Identify the leaving variable. Compare both eligible ratios, perform the pivot, then decide whether another pivot is required.

Check the basic values and the reduced costs separately. The next page gives the complete answer.

Practice answer: both feasibility tests pass

x_4 leaves. Compare $3/1 = 3$ and $2/4 = 1/2$, so x_2 enters. Solving and substituting gives

$$x_2 = \frac{1}{2} - \frac{1}{4}x_1 + \frac{1}{4}x_4,$$

$$x_3 = 3 + \frac{3}{2}x_1 - \frac{1}{2}x_4,$$

$$f = 1 + \frac{5}{2}x_1 + \frac{1}{2}x_4.$$

Set $x_1 = x_4 = 0$: $x_2 = 1/2$, $x_3 = 3$, and $f = 1$. All basic values and reduced costs are nonnegative, so the basis is optimal.

A zero reduced cost can give a zero-improvement pivot

If the entering reduced cost is zero, the dual ratio is zero. The pivot can change the basis without changing its dual objective value.

For a dual basic solution in $\mathbb{R}^m$, the m basic-column constraints are active. An additional active constraint means dual degeneracy.

Nonbasic $\bar{c}_j = 0 \iff$ an additional dual constraint is active.

This is different from a zero basic primal variable. Degeneracy belongs to the primal or dual solution being discussed.

Prevent cycling: lexicographic comparison of columns

Use the tableau convention already introduced. Keep the objective row first, followed by a fixed constraint-row order.

Choose any negative RHS row ℓ . For each eligible column j , divide its **entire column** by $|v_j|$, where $v_j < 0$ is its entry in row ℓ .

Compare the normalized columns from top to bottom. Choose the lexicographically smallest; break a complete tie by the smallest variable index.

If every initial column is lexicographically positive, this rule terminates after finitely many pivots. Lexicographically positive means its first nonzero entry is positive.

The lexicographic test extends the ordinary ratio test

In Example 4.7, the eligible initial columns are 2 and 3. Read each column as objective row, first constraint row, second constraint row.

$$\frac{C_2}{2} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, \quad \frac{C_3}{3} = \begin{pmatrix} 10/3 \\ 1/3 \\ -1 \end{pmatrix}.$$

The first entries already differ: $3 < 10/3$. Thus column 2 enters. If the ratios tied, the next unequal entry would decide.

All initial columns here are lexicographically positive. This is a termination condition, not an extra eligibility test on the pivot row.

Example 4.9: one primal point can represent several bases

$$\min f = 3x_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2, \quad 2x_1 - x_2 \geq 0, \quad x_1, x_2 \geq 0.$$

$$\max 2p_1 \quad \text{s.t.} \quad p_1 + 2p_2 \leq 3, \quad p_1 - p_2 \leq 1, \quad p_1, p_2 \geq 0.$$

Add surplus variables: $x_1 + x_2 - x_3 = 2$, $2x_1 - x_2 - x_4 = 0$.

At $(x_1, x_2) = (0, 0)$, the full basic solution is $(0, 0, -2, 0)$. Its basic zero causes different bases to represent the same primal point.

The associated dual solution depends on the basis

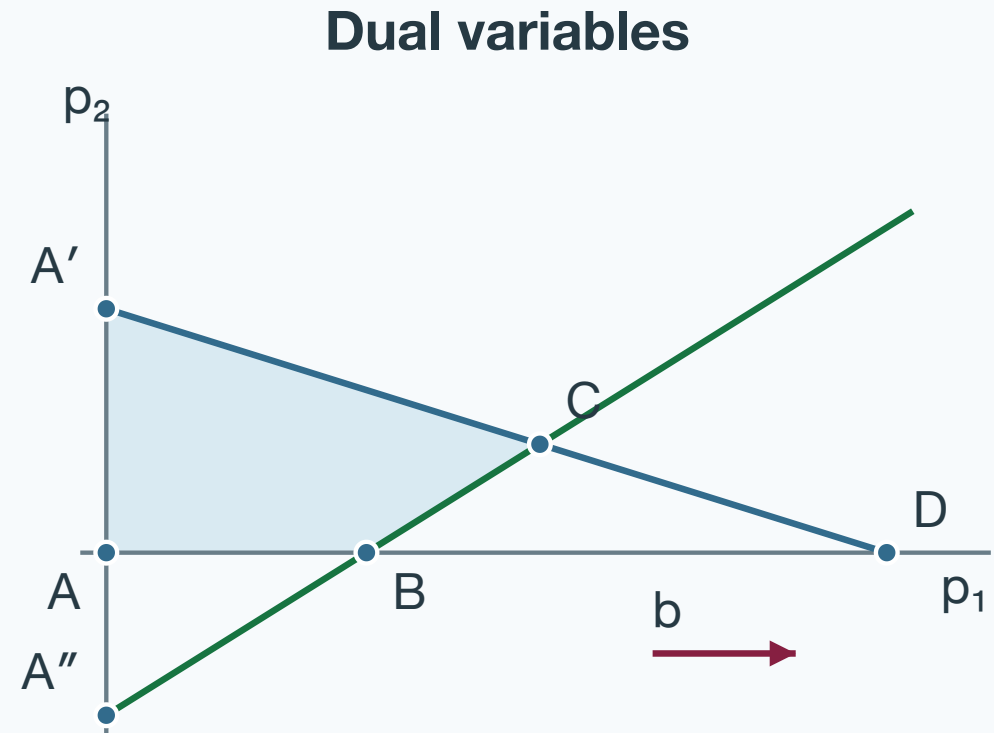
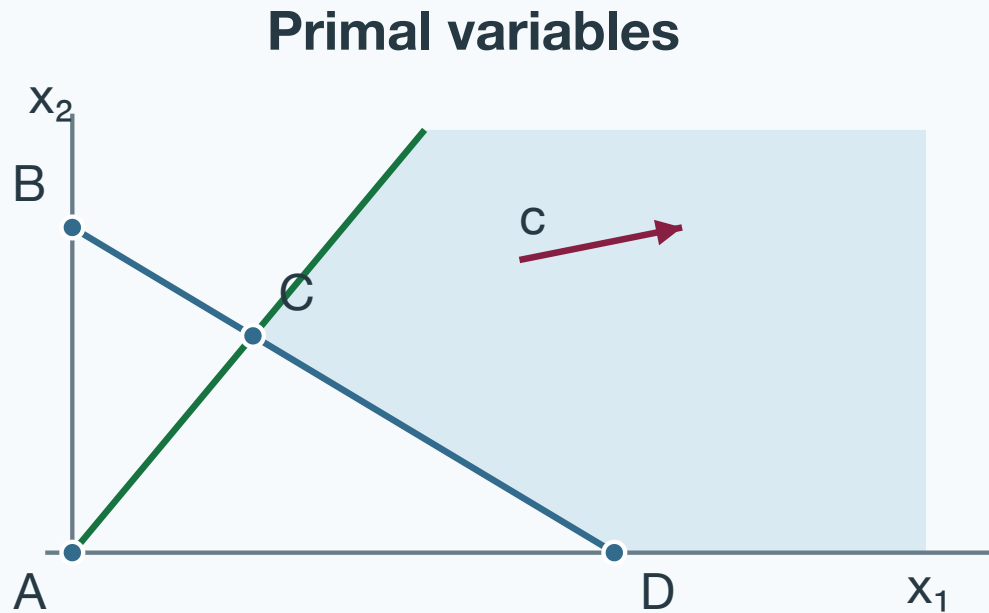
Ordered basic variables	Active dual equalities	Associated p
(x_3, x_4)	$-p_1 = 0, \quad -p_2 = 0$	$(0, 0)$: feasible
(x_1, x_3)	$p_1 + 2p_2 = 3, \quad -p_1 = 0$	$(0, 3/2)$: feasible
(x_2, x_3)	$p_1 - p_2 = 1, \quad -p_1 = 0$	$(0, -1)$: infeasible

All three bases give the same primal basic solution. They give different dual basic solutions, and one fails dual feasibility.

Therefore, inspect the reduced costs of the particular basis; the primal point alone does not determine them.

Three dual points correspond to the same primal point A

Example 4.9: $\min 3x_1 + x_2$, with $x_1 + x_2 \geq 2$, $2x_1 - x_2 \geq 0$, $x \geq 0$. The dual maximizes $2p_1$, with $p_1 + 2p_2 \leq 3$, $p_1 - p_2 \leq 1$, $p \geq 0$.



The same primal point can have different associated dual points. Maroon arrows show increasing objective; blue shading is feasible.

From an impossible row to a general infeasibility certificate

Dual simplex stops successfully when both feasibility tests hold. It detects infeasibility when a negative row has no eligible entering variable.

That impossible row is a linear combination of the original equations. It proves that every nonnegative candidate must fail.

Next, Farkas' lemma answers the broader question: does every infeasible system have such a short certificate?

Study guide: marginal costs and dual simplex

Be able to do	Where it comes from
Interpret a dual variable and explain its validity range	§4.4
Classify a dictionary as primal feasible, dual feasible, both or neither	§4.5
Perform a dual-simplex pivot and verify both tests	Examples 4.7–4.8
Explain the paired geometry and degeneracy caveat	Examples 4.8–4.9
Recognize the two stopping cases and the anticycling condition	§4.5

The appendix contains full calculations, proofs, and further examples. All principal algorithm rules and stopping conditions are in the main sequence.

Appendix: calculations, proofs, and further examples

The main sequence states every pivot rule and stopping condition. These pages verify the algebra and preserve full proofs.

Marginal costs and validity ranges

Dual-simplex correctness and stopping cases

Lexicographic finite termination

Worked tableaux and basis geometry

Calculate the range from nonnegative quantities

Keep $x_2 = 0$, the first target 8, and the second target $3 + \varepsilon$.

$$x_1 = 1 + \frac{\varepsilon}{3} \geq 0 \iff \varepsilon \geq -3,$$

$$x_3 = 1 - \frac{5\varepsilon}{9} \geq 0 \iff \varepsilon \leq \frac{9}{5}.$$

Thus the current basis is feasible for $-3 \leq \varepsilon \leq 9/5$. The reduced costs stay $(0, 7, 0)$, so this basis stays optimal throughout the interval.

At an endpoint a basic quantity is zero. At an interior point both basic quantities are positive.

Beyond the range, a new basis gives a new price

With target $(8, 5)$, set $x_3 = 0$ and solve

$$5x_1 + x_2 = 8, \quad 3x_1 + x_2 = 5 \quad \Rightarrow \quad x = (3/2, 1/2, 0).$$

$$c^\top x = \frac{49}{2} = 24.5.$$

Active dual constraints for the positive variables give $5p_1 + 3p_2 = 13$ and $p_1 + p_2 = 10$, hence $p = (-17/2, 37/2)$.

The remaining inequality is $3p_1 = -51/2 \leq 6$; the dual value is $8(-17/2) + 5(37/2) = 49/2$. This certifies the new optimum.

The arithmetic has a compact matrix form

For an optimal basis B with $B^{-1}b \geq 0$ and nonnegative reduced costs, the quantities and dual variables are

$$x_{\mathcal{B}} = B^{-1}b, \quad (p^*)^\top = c_{\mathcal{B}}^\top B^{-1}.$$

$$\begin{aligned} c_{\mathcal{B}}^\top B^{-1}(b + \Delta b) &= c_{\mathcal{B}}^\top B^{-1}b + c_{\mathcal{B}}^\top B^{-1}\Delta b \\ &= v(b) + (p^*)^\top \Delta b. \end{aligned}$$

This is the optimum whenever $B^{-1}(b + \Delta b) \geq 0$, because the reduced costs do not depend on b .

Why nondegeneracy gives room for small changes

Assume $B^{-1}b > 0$. Every current basic quantity is strictly positive.

The changed quantities are $B^{-1}b + B^{-1}\Delta b$. For sufficiently small Δb , none can reach zero.

Thus the same basis stays feasible in a neighborhood, and the value is a linear function there. Its coefficient on requirement i is the optimal dual variable for that row.

If a basic quantity already equals zero, some arbitrarily small changes may make it negative. This explains why the nondegeneracy assumption matters for a two-sided neighborhood.

Why the minimum cost is piecewise linear and convex

With A, c fixed, let $D = \{p : A^\top p \leq c\}$. At every feasible RHS with finite optimum, strong duality gives

$$v(b) = \max_{p \in D} b^\top p.$$

Each feasible dual solution supplies a linear lower bound. Their maximum is a convex, piecewise-linear value function on the region with finite value.

Inside a region where one optimal basis remains feasible, its formula is linear. At a boundary, another dual solution can determine a different slope.

A finite value does not imply differentiability. An optimal dual vector remains a valid supporting lower bound even at a corner.

Proof: the unchanged basis gives the new optimum

Assume the basis B has $\bar{c} \geq 0$, and let $\hat{b} = b + \Delta b$ satisfy $B^{-1}\hat{b} \geq 0$.

$$\hat{x}_{\mathcal{B}} = B^{-1}\hat{b}, \quad \hat{x}_{\mathcal{N}} = 0, \quad p^{\top} = c_{\mathcal{B}}^{\top}B^{-1}.$$

The first vector is primal feasible. Because A, c, B have not changed, $A^{\top}p \leq c$ remains true.

$$c^{\top}\hat{x} = c_{\mathcal{B}}^{\top}B^{-1}\hat{b} = p^{\top}\hat{b}.$$

Weak duality and equal objective values prove that both vectors are optimal. Subtract the corresponding old equality to obtain the marginal-cost formula.

Proof: substitute the pivot equation into the objective

Write the negative row as $x_k = \beta + \sum_i \alpha_i x_i$, with $\beta < 0$. Choose $\alpha_j > 0$. Solving gives

$$x_j = -\frac{\beta}{\alpha_j} + \frac{x_k}{\alpha_j} - \sum_{i \neq j} \frac{\alpha_i}{\alpha_j} x_i.$$

Substitute into $f = f_0 + \sum_i \bar{c}_i x_i$, and let $t = \bar{c}_j / \alpha_j$:

$$f = f_0 - t\beta + tx_k + \sum_{i \neq j} (\bar{c}_i - t\alpha_i) x_i.$$

The minimum ratio keeps all displayed coefficients nonnegative. This is dual feasibility in the new basis.

Proof: the minimum ratio covers every remaining column

Let $t = \min_{\alpha_i > 0} \bar{c}_i / \alpha_i \geq 0$. For every remaining nonbasic variable:

$$\begin{aligned}\alpha_i > 0 &\implies t \leq \bar{c}_i / \alpha_i \implies \bar{c}_i - t\alpha_i \geq 0, \\ \alpha_i \leq 0 &\implies \bar{c}_i - t\alpha_i \geq \bar{c}_i \geq 0.\end{aligned}$$

The leaving variable becomes nonbasic with coefficient $t \geq 0$. The entering variable becomes basic with reduced cost zero.

Thus every dual constraint still holds after the pivot. Feasibility of other primal basic values is not asserted; they must be checked at the next iteration.

Proof: positive ratios give strict dual improvement

$$f'_0 = f_0 - t\beta, \quad \beta < 0, \quad t \geq 0.$$

If $\bar{c}_j > 0$, then $t > 0$, so $f'_0 > f_0$. This is an increase of the associated dual objective.

There are finitely many bases. A strictly increasing objective prevents a basis from repeating, so the method must terminate.

At termination, either all primal basic values are nonnegative and the pair is optimal, or a negative row has no eligible entering column.

If the ratio is zero, strict improvement fails. This is why a separate anticycling rule is needed in degenerate cases.

Proof: an impossible row gives a certificate and a dual ray

Let ℓ be the selected row. Write $q^\top = e_\ell^\top B^{-1}$, where e_ℓ selects row ℓ . If all tableau entries in that row are nonnegative,

$$q^\top A \geq 0, \quad q^\top b = \beta < 0.$$

A feasible $x \geq 0$ would give $0 \leq q^\top Ax = q^\top b < 0$, a contradiction.

$$p(t) = p - tq : \quad A^\top p(t) \leq c, \quad b^\top p(t) = b^\top p - t\beta \longrightarrow +\infty.$$

Here $t \geq 0$ and p is the current feasible dual solution. Thus the primal is infeasible and the dual is unbounded.

Lexicographic proof: track full tableau columns

Let $C_j = (\bar{c}_j, u_{1j}, \dots, u_{mj})^\top$, with objective row first and fixed constraint-row positions. Assume every initial C_j is lexicographically positive.

Choose row ℓ , with RHS $\beta < 0$, and let $v_j = u_{\ell j} < 0$ be the entering entry. Since C_j is lex positive but $v_j < 0$, its first nonzero entry occurs **before** row ℓ .

$$\frac{C_j}{|v_j|} \leq_{\text{lex}} \frac{C_i}{|v_i|} \quad \text{for every eligible } v_i < 0.$$

We prove that all columns remain lex positive and that the full RHS column strictly decreases lexicographically.

Lexicographic proof: columns with nonnegative pivot-row entries

For another column i with $v_i \geq 0$, subtracting the pivot-column multiple gives, outside row ℓ ,

$$C_i - \frac{v_i}{v_j} C_j = C_i + \frac{v_i}{|v_j|} C_j.$$

If $v_i > 0$, its first nonzero entry before row ℓ is positive: both columns are lex positive, and C_j already has a positive first nonzero entry before that row.

Changing the pivot-row entry to v_i/v_j cannot change that earlier sign. If $v_i = 0$, the column is unchanged.

These columns therefore stay lexicographically positive.

Lexicographic proof: eligible columns also stay positive

For another eligible column i , the chosen lexicographic minimum gives

$$D_i = C_i - \frac{|v_i|}{|v_j|} C_j \geq_{\text{lex}} 0.$$

Its entry in row ℓ is zero. The actual pivot replaces that entry by $v_i/v_j > 0$.

If the first nonzero entry of this difference is before the pivot row, it is positive. Otherwise, the positive replacement in the pivot row becomes its first nonzero entry.

The entering column itself becomes e_ℓ , also lex positive. This proves preservation for every column.

Lexicographic proof: no ordered basis can repeat

Let $C_0 = (-f_0, \bar{b}_1, \dots, \bar{b}_m)^\top$ be the full RHS column. Outside the pivot row,

$$C'_0 = C_0 - \frac{\beta}{v_j} C_j, \quad \frac{\beta}{v_j} > 0.$$

The first nonzero entry of C_j is positive and occurs before row ℓ . Hence the first changed entry of C_0 decreases. Normalizing the later pivot-row entry cannot undo this.

$$C'_0 <_{\text{lex}} C_0.$$

Every ordered basis determines one tableau, and there are finitely many ordered bases. Strict lexicographic decrease rules out repetition. Therefore the method terminates.

Example 4.8 in tableau form: initial state

$$\min f = x_1 + x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \quad x_1, x_2 \geq 0.$$

Row	Column 0	x_1	x_2	x_3	x_4
R_0	0	1	1	0	0
x_3	-2	-1	-2	1	0
x_4	-1	-1	0	0	1

Column 0 is the RHS in constraint rows and $-f_0$ in R_0 , where $f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j$. Columns always follow natural variable order.

The negative row $x_3 = -2$ leaves; the ratios are 1 and $1/2$. Column x_2 enters.

Example 4.8 in tableau form: after pivot 1

Row	Column 0	x_1	x_2	x_3	x_4
R_0	-1	1/2	0	1/2	0
x_2	1	1/2	1	-1/2	0
x_4	-1	-1	0	0	1

$$\begin{aligned}x_2 &= 1 - \frac{1}{2}x_1 + \frac{1}{2}x_3, \\x_4 &= -1 + x_1, \\f &= 1 + \frac{1}{2}x_1 + \frac{1}{2}x_3.\end{aligned}$$

The new objective is 1, but $x_4 = -1$ remains infeasible. Column x_1 is the only eligible entering column for that row.

Example 4.8 in tableau form: after pivot 2

Row	Column 0	x_1	x_2	x_3	x_4
R_0	$-3/2$	0	0	$1/2$	$1/2$
x_1	1	1	0	0	-1
x_2	$1/2$	0	1	$-1/2$	$1/2$

$$\begin{aligned}x_1 &= 1 + x_4, \\x_2 &= \frac{1}{2} + \frac{1}{2}x_3 - \frac{1}{2}x_4, \\f &= \frac{3}{2} + \frac{1}{2}x_3 + \frac{1}{2}x_4.\end{aligned}$$

All basic values and reduced costs are nonnegative; $f^* = 3/2$. The constraint rows above are reordered only for this final display; lexicographic iteration must keep fixed row positions.

Why each basis gives a basic dual solution

A basis consists of m independent columns $A_{B(1)}, \dots, A_{B(m)}$. The associated dual solution satisfies

$$A_{B(i)}^\top p = c_{B(i)}, \quad i = 1, \dots, m.$$

These are m independent active equalities in the m -dimensional variable p . Their unique intersection is therefore a basic dual solution.

It is feasible exactly when all remaining inequalities $A_j^\top p \leq c_j$ also hold—equivalently, when all reduced costs are nonnegative.

A pivot changes one basic column, so consecutive feasible dual bases share $m-1$ independent active equalities. Their points are adjacent or coincide.

Example 4.9: all six bases

Basic variables	Primal (x_1, x_2)	Dual (p_1, p_2)
(x_3, x_4)	$(0, 0)$	$(0, 0)$
(x_1, x_3)	$(0, 0)$	$(0, 3/2)$
(x_2, x_3)	$(0, 0)$	$(0, -1)$
(x_2, x_4)	$(0, 2)$	$(1, 0)$
(x_1, x_2)	$(2/3, 4/3)$	$(5/3, 2/3)$
(x_1, x_4)	$(2, 0)$	$(3, 0)$

Every row comes from setting nonbasic primal variables to zero and setting basic-column dual constraints active. Distinct bases need not mean distinct primal points.

Why a zero nonbasic reduced cost means dual degeneracy

At the dual point associated with B , the m basic-column constraints are active. For any other column,

$$\bar{c}_j = c_j - A_j^\top p = 0 \quad \Longleftrightarrow \quad A_j^\top p = c_j.$$

Thus a zero nonbasic reduced cost supplies an additional active constraint beyond the m constraints defining the basis.

This can produce multiple dual bases at one dual point. Conversely, every additional active nonbasic-column constraint has reduced cost zero.

This observation concerns dual degeneracy; a zero primal basic value is a separate condition.

Which property survived the model change?

Change from an optimal basis	Property preserved	Natural next step
Only RHS b changes	Dual feasibility	Check basic values; use dual simplex if needed.
Only costs c change	Primal feasibility	Check reduced costs; use primal simplex if needed.
A new nonnegative variable is added	Old primal point extends with zero	Check the new reduced cost.
A new $\leq$ constraint is added	Old dual solution extends with zero	Check the new slack; dual simplex if negative.

This is a starting-state diagnosis, not a guarantee of fewer pivots. Sensitivity analysis will calculate the ranges where no pivot is needed.